

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date: 18 June 2026

Team ID: 24EG105H29

Project Name: AuraCare – Smart Healthcare & Telemedicine Management System

Maximum Marks: 4 Marks

Technical Architecture

Patient/Doctor/Admin → React.js Frontend → Express.js APIs → Node.js Business Logic → MongoDB Database. JWT Authentication secures access to appointments, telemedicine, prescriptions, and emergency services.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web-based healthcare portal	HTML, CSS, JavaScript, React.js
2	Application Logic-1	Authentication & User Management	Node.js, Express.js
3	Application Logic-2	Appointment & Telemedicine Management	Node.js, Express.js
4	Application Logic-3	Prescription & Emergency Services	Node.js, Express.js
5	Database	Patient, Doctor, Hospital & Appointment Data	MongoDB
6	Cloud Database	Database Hosting	MongoDB Atlas / MongoDB
7	File Storage	Medical Reports & Documents	Local Storage / MongoDB
8	External API-1	Video Consultation Integration	WebRTC / Browser APIs
9	External API-2	Emergency Service Integration	Emergency Hotline Services
10	Machine Learning Model	Not Implemented	N/A
11	Infrastructure	Application Deployment	Local System, VS Code, Node.js Server

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used in development	React.js, Node.js, Express.js, MongoDB
2	Security Implementations	Authentication and authorization	JWT Authentication, Password Hashing
3	Scalable Architecture	Supports increasing users and hospitals	MERN Stack Architecture
4	Availability	Accessible anytime through web application	24/7 Web Access
5	Performance	Fast response and optimized queries	MongoDB, REST APIs